

Metali Mangal

Research Interests

Human-AI Collaboration, Computer Graphics, Neural Rendering, Extended Reality (XR), Games and Interactive Systems, Human-Computer Interaction.

Education

Purdue University, West Lafayette, IN *Jan 2024 – Present*

Ph.D. in Computer Graphics and Technology (Games, AR/VR)

- Research Focus: Human-AI Collaboration, Computer Graphics, Neural Rendering, and Extended Reality training systems.
- Developing VR-based AI-guided training environments and conducting studies on user engagement and learning outcomes.

Purdue University, West Lafayette, IN *Aug 2021 – Aug 2023*

M.S. in Computer Graphics and Technology (Games, AR/VR)

Thesis: Gamification of Virtual Reality Training – Effects of Visual Motivation

- Studied the impact of game elements on VR micro-pipette training.
- Selected courses: Human-Computer Interaction, Computer Graphics, Interactive Systems, VR Development, AI.

Manipal University, Manipal, India *Aug 2013 – Jun 2017*

B.Tech in Computer Science and Engineering

- Award: Scholarship for Top 1000 Rank in Entrance Exam.
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Publications & Research Contributions

- **“Inclusion Through Game Design: A Non-Serious Game Focused on Inclusion and Representation for Children with Physical Disabilities,”** submitted to CHI PLAY, 2026 – Third author and developer. Explored design strategies to improve accessibility and inclusion in educational games.
 - **Ongoing Research: “I am Innocent: How Emotional Persuasion Influences Suspects’ Judgements in a Murder-Mystery Interrogation with Three Virtual Agents.”** Studying the effects of emotion-driven AI agents on decision-making and trust in interactive simulations.
 - **Ongoing Research: “Gaussian Human Avatars and their perception”** Investigating uncanny valley effects in Gaussian-rendered human avatars, and how varying levels of visual realism influence perceived naturalness and comfort in interactive simulations.
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Professional Experience

Teaching Assistant, Purdue University *Aug 2024 - Present*

- Teach Introduction to Web Development, guiding students in HTML, CSS, and JavaScript.
- Mentor students on coding best practices, collaboration, and project workflows.
- Introduced industry-standard tools (GitHub, Kanban boards, Jira) to prepare students for real-world development.

Simulation Developer, Purdue Envision Center *Jan 2022 - Aug 2023*

West Lafayette, IN

- Developed immersive VR simulations for laboratory training using Unity, WebXR, and WebGL.
- Led projects including “BIO-Spinach PE” and “CNC Machine,” improving student training engagement and efficiency.
- Collaborated across departments to align technical implementations with educational objectives.

Unity Developer, Iconic Engine *Aug 2022 - Dec 2022*

- Contributed to Sports Metaverse Project; led development of collaboration features in multiplayer VR environments.

Software Developer, Games 24x7 Pvt. Ltd. *Feb 2019 - Jul 2021*

Bengaluru, India

- Enhanced iOS game testing efficiency by 30% via a unified game-table automation framework.
- Managed CI/CD pipeline for automated builds using Jenkins.
- Awarded Extra Miles Award for contributions to the iOS Automation framework.

Software Developer, Sling (Dish Networks)

Oct 2017 - Jan 2019

Bengaluru, India

- Designed and developed an iOS automation framework integrated with IoT devices.
 - Awarded Spot Award for excellence in framework development.
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Projects

- **Social VR Amusement Park (VR)** – Led multiplayer VR development for social interaction.
 - **VR Basketball Simulation** – Coordinated team and stakeholder expectations for a functional VR sports experience.
 - **Unity VR Tutorial** – Mentored peers and undergraduates on creating VR-based learning environments.
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Technical Skills

- **Programming:** C#, C++, Java, Python, Swift, HTML5/CSS, GLSL
 - **Game/ Tools:** Unity, Unreal Engine, Blender, ARCore, OpenXR, Photon, OpenGL
 - **Tools & Workflow:** Git, Jenkins, Azure, Perforce, CI/CD, Appium, JIRA
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Awards & Honors

- Extra Miles Award, Games 24x7 (iOS Automation Framework)
- Spot Award, Sling (iOS Automation Framework with IoT Integration)